

Qatar-1b

R-0526

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_190812 · created 2026-07-21T04:08:57+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458707.7829 +/- 0.0016 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.088 +/- 0.054
Transit depth (Rp/Rs)^2	0.77 +/- 0.95 [%]
Orbital Inclination (inc)	79.82 +/- 2.93
Airmass coefficient 1 (a1)	0.065 +/- 0.019
Airmass coefficient 2 (a2)	2.22 +/- 0.89
Scatter in the residuals of the lightcurve fit is	28.77 %
Best Comparison Star	None
Optimal Aperture	3.0
Optimal Annulus	7.5
Transit Duration (day)	0.021 +/- 0.025

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.088 $\pm$ 0.054 vs 0.1463 (deviation 1.08 σ)
Duration fit vs published	0.021 d vs 0.0692 d (deviation 1.93 σ)
Mid-time O-C	-4.0 min
Depth SNR	1.6
Residual scatter	28.77 %

Light curve

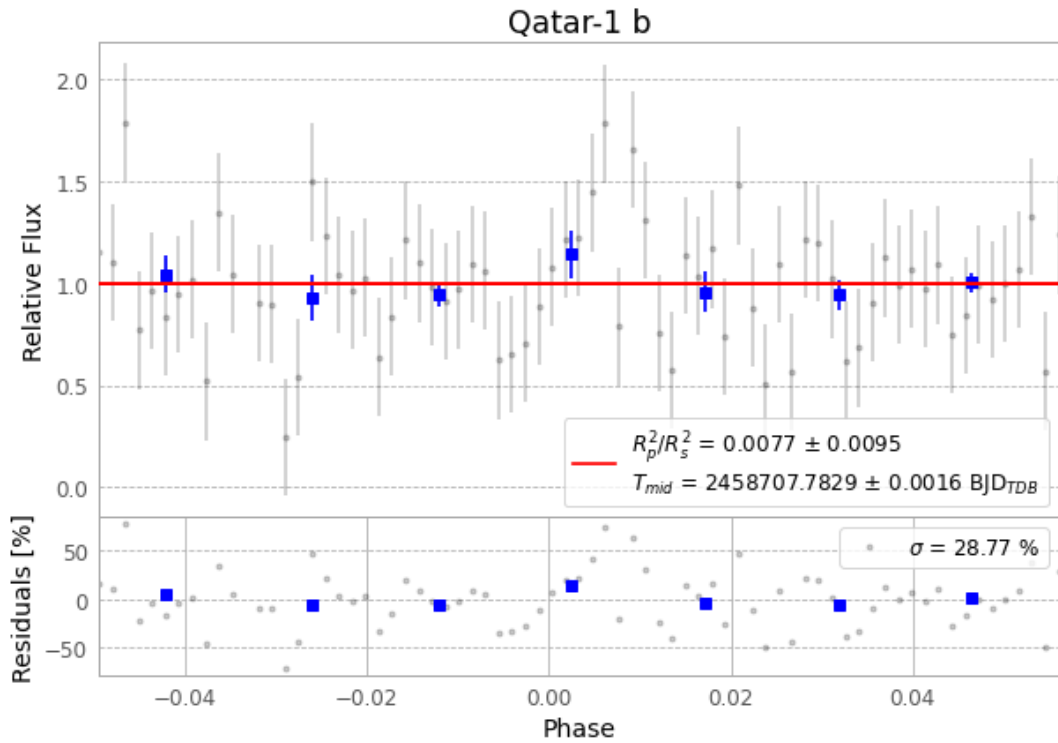


Figure 1 — FinalLightCurve_Qatar-1 b_2019-08-11.png (SHA-256 2bcb841188c36829...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [313, 234], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 43728b27776ea56c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 72f1d3a

Data provenance

73 FITS frames (48 MB), Qatar-1190812050315.FITS → Qatar-1190812083912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-190812.log (e1e40f6257da...), FinalLightCurve_Qatar-1 b_2019-08-11.png (2bcb841188c3...), FinalLightCurve_Qatar-1 b_2019-08-11.csv (f6ef7688ad84...)

Compute resources

Wall time 110.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-21 04:08Z.